

End-to-End Modelling of Earthquake-Induced Polarisation Perturbations in Submarine Optical Fibres

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Abstract We present an end-to-end model of earthquake-perturbed signal propagation in submarine fibres. Ground displacements are converted to fibre strain via ocean pressure, and injected into a split-step model. Our model shows good qualitative alignment with measurements from the Curie cable during the 2020 Oaxaca earthquake. ©2026 The Author(s)

Introduction

Earthquake early warning systems [1] traditionally rely on seismometer networks [2]. However, off-shore seismometer presence is sparse, whereas optical communication fibres are widespread [3]. Therefore, it has been proposed to use existing submarine telecom fibres for earthquake sensing [4–6]. Indeed, earthquakes induce spatiotemporal perturbations of the fibre geometry [7]. Fibre strain affects the birefringence, i.e., the polarisation-dependence of the speed of light in the fibre. Hence, perturbations can be sensed from variations in the received state of polarisation (SOP).

Sensing algorithm design requires earthquake-perturbed fibre output data. Capturing the resulting SOP variations in measurements is challenging [8], and only a few experimental datasets are publicly available, e.g., [9, 10]. Alternatively, given an earthquake event, fibre link, and input signal, a numerical propagation model could provide controlled and reproducible fibre output SOP traces. Such a model would support the development, testing, and comparison of sensing algorithms without requiring rare measurement data.

To the best of our knowledge, only one such propagation model has been previously proposed, which assumes buried terrestrial fibres [11, 12]. This model obtains earthquake seismograms from a web service that synthesises seismograms (Syn-gine [13]), and integrates them into a numerical waveplate model that solves the Manakov-PMD (polarisation mode dispersion) equation [14]. However, the Manakov-PMD equation was derived in a reference frame that follows the rapid birefringence-induced SOP rotations at the carrier [15]. Therefore, strictly speaking the model in [12, App. A] predicts no SOP perturbations at the carrier frequency $\omega = 0$ during earthquakes.¹ Moreover, the model assumes a linear relation

between ground motion and fibre strain, whereas submarine environments have a more complex transfer function [16, 17].

In this paper, we present an end-to-end simulation model of the full interaction chain from earthquake-induced ground motion to optical signal propagation in submarine fibres. To the best of our knowledge, this is the first such model, and we demonstrate good qualitative alignment with measurements from [9]. In contrast to [11, 12], our model is based on the coupled Schrödinger equation (CSE) which resolves deterministic, time-dependent perturbations at the carrier. Moreover, our implementation is open source², enabling reproducible sensing research and simulation of earthquake-induced fibre propagation without requiring experimental data.

Earthquake-perturbed propagation model

In the linear regime, the CSE describes how an optical carrier wave propagates through single-mode fibre according to [18, Ch. 2] and [15]

$$j \frac{\partial \vec{A}(z, t')}{\partial z} + b \Sigma(\theta(z)) \vec{A}(z, t') = 0 \quad (1)$$

with $j = \sqrt{-1}$, dual-polarisation complex signal envelope $\vec{A}(z, t') \in \mathbb{C}^2$, position z , time t' relative to group velocity, half birefringence $b = \pi/L_{\text{beat}}$ at the carrier frequency, and beat length L_{beat} . In (1), $\Sigma(\theta(z)) = \begin{bmatrix} \cos 2\theta(z) & \sin 2\theta(z) \\ \sin 2\theta(z) & -\cos 2\theta(z) \end{bmatrix}$ ensures that the birefringence is applied between the fastest- and slowest-travelling polarisations, where the fastest makes an angle $\theta(z)$ with a fixed reference frame. In long-haul fibre systems, amplifiers add noise to the envelope during propagation. We chose to omit attenuation and amplification, and model noise only at the end of the fibre, as we will describe in the next section. As $\theta(z)$ is a stochastic process in z , (1) has no closed-form solution.

¹The authors of [14] likely evaluated their model at an offset frequency $\omega \neq 0$.

²<https://github.com/RM-8vt13r/quake-fibre>

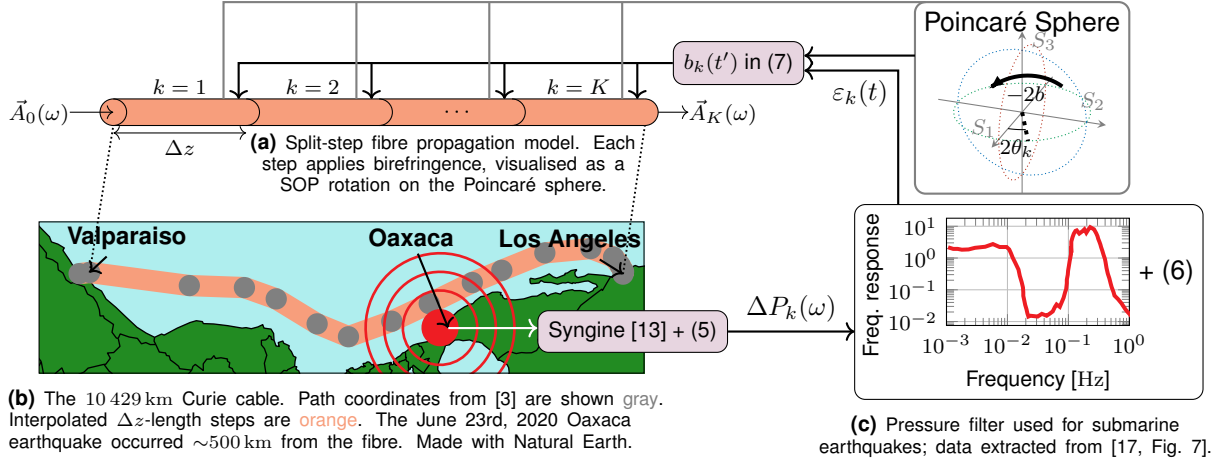


Fig. 1: Block diagram of the end-to-end earthquake-perturbed propagation model.

The SOP is traditionally written as normalised real Stokes vector $\vec{S} = [S_1 \ S_2 \ S_3]^T$ on the Poincaré sphere, as visualised in Fig. 1a (right).

To evaluate (1) numerically, we follow [15] and divide a fibre into K Δz -length steps as visualised in Fig. 1a (left). For each step $k \in \{1, \dots, K\}$, we generate $\theta_k = \theta(k\Delta z)$ as

$$\theta_k - \theta_{k-1} \sim \mathcal{N}\left(0, \frac{\Delta z}{2L_{\text{corr}}}\right), \quad (2)$$

with correlation length L_{corr} and $\theta_0 = 0$. Similar to [15, Eq. 23], we evaluate step k as

$$\vec{A}_k(\omega) = R_k D R_k^{-1} \vec{A}_{k-1}(\omega), \quad (3)$$

$$R_k = \begin{bmatrix} \cos \theta_k & -\sin \theta_k \\ \sin \theta_k & \cos \theta_k \end{bmatrix}, D = e^{j b \Delta z} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad (4)$$

where $\vec{A}_k(t') = \vec{A}(k\Delta z, t')$, ω is frequency relative to the carrier and e° is the Hadamard exponential.

Now, we add the effect of earthquakes in (3) as shown schematically in Fig. 1. First, we divide a fibre path [3] into steps of length Δz using linear spline interpolation, as Fig. 1b shows. Given measurements of a historic or hypothetical earthquake, Syngine [13] synthesises ground displacements in the centre of step k . We retrieve historical earthquake measurements from the Global Centroid Moment Tensor database [19]. Vertical ground displacement $u_k(t)$ at fibre step k , with absolute time t , creates water pressure differences [20]

$$\Delta P_k(t) = \rho h \frac{\partial^2 u_k(t)}{\partial t^2}, \quad (5)$$

with water depth h , and water density ρ . We assume h and ρ constant.

The expression in (5) was derived assuming that ocean water is incompressible. In practice, this assumption breaks down above the h -dependent ocean resonance frequency [21]. [17] measured broadband seafloor pressure during seismic activity, demonstrating this frequency-dependence. We assume that their pressure measurements represent the frequency response of the water column.

Tab. 1: Curie model parameters. We took ξ and ϵ_P from [7], L_{beat} , L_{corr} and Δz from [15], and ρ and h from [22, Fig. 3].

Symbol	Value	Symbol	Value
β_1	2.1×10^8 m/s	$K\Delta z$	10 429 km
L_{beat}	50 m	ρ	1045 kg/m
L_{corr}	100 m	h	4 km
ξ	0.78	ϵ_P	3.5/GPa
Δz	1.67 m		

Therefore, we propose to model this frequency response by filtering $\Delta P_k(t)$ with the filter in Fig. 1c, taken from [17, Fig. 7b]. Thus, our model relies on experimental measurements of the ocean frequency response.

Finally, we obtain fibre strain [7, Secs. 3 and 4]

$$\varepsilon_k(t) = \alpha \xi \epsilon_P \Delta P_k(t), \quad (6)$$

where $\Delta P_k(t)$ has been filtered as described, photoelasticity ξ corrects for refractive index changes due to material compression, and ϵ_P scales pressure to strain. Here, we introduced a calibration factor α to compensate uncertainties in h , ρ , ϵ_P , and the pressure filter. We assume that $\varepsilon_k(t)$ is stepwise constant in t and replace b in (4) by [7]

$$b_k(t') = (1 + \varepsilon_k(t' + \beta_1 k \Delta z)) b, \quad (7)$$

where t' is matched to t using the polarisation-averaged group velocity β_1 [18, Eq. 2.3.44].

Simulation, results & discussion

We reproduce the experiment from [9] in simulation with the setup from Fig. 1 and Tab. 1, transmitting a dual-polarisation carrier wave at $\omega = 0$ (wavelength 1550 nm). We chose a small value of Δz similar to [15] to capture rapid birefringence variations, resulting in a simulation time of roughly 6 h on an NVIDIA Tesla V100 GPU. We set $\alpha = 1.52$ and add Gaussian noise to $\vec{A}(K\Delta z, t')$ to match the measured data as closely as possible, and then convert to Stokes parameters³.

³Note that [9] estimates the Stokes parameters from an equaliser in the receiver, instead of from $\vec{A}(K\Delta z, t')$ as we do.

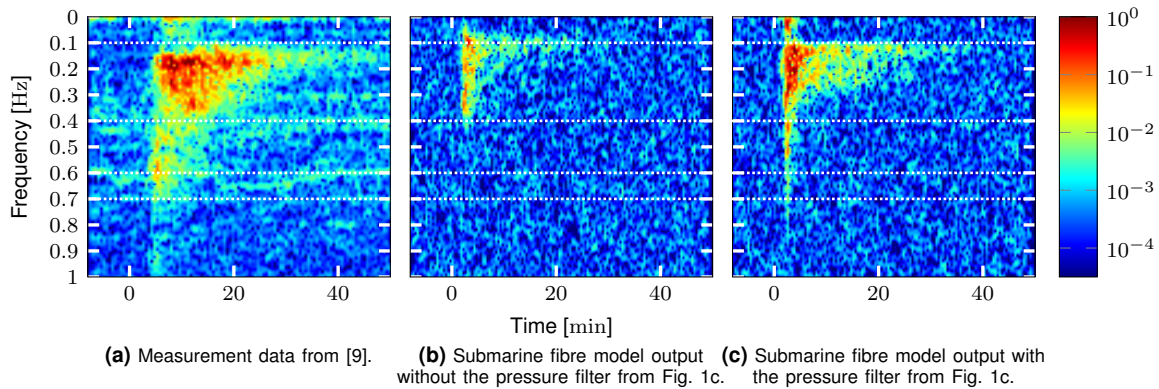


Fig. 2: Measured and modelled spectrograms, calculated over S_1 and S_2 and then summed. The earthquake starts at 0 min.

Like [9], we align the SOP in Stokes space to the average S_3 over a 200 s moving average, which essentially acts as a high-pass filter. Then, to visualise the effect of the earthquake on the fibre output, we follow [9] and sum the spectrograms of S_1 and S_2 . We generate spectrograms using a 102.4 s Blackman-Harris window, 75 % window overlap and 409.6 s zeropadded Fourier transforms. The result is shown in Fig. 2, where Fig. 2a shows the result from [9], and Fig. 2b and Fig. 2c show our model output excluding- and including the pressure filter from Fig. 1c. The spectrograms visualise the amplitude of the fibre output SOP rotations, as a function of time and frequency. The results shown are for one random realisation of θ_k . We also verified that different realizations produce very similar spectrograms, with only a minor amplitude change near 0 min.

Both in the measurements and our filtered model, the SOP fluctuates in the 0 Hz to 0.7 Hz band when the earthquake begins near 0 min. For the unfiltered model, this is 0.05 Hz to 0.4 Hz. In both the measurement and our filtered model, the SOP rotations are strongest between 0.1 Hz and 0.4 Hz. In this range, the model output shows larger spectrogram amplitudes than the experimental measurements. This amplitude is prone to our choice of α , and the strong pressure amplification by the filter from Fig. 1c in this band.

After the earthquake reaches the fibre, the SOP fluctuations diminish over time in all spectrograms. Here, high-frequency components attenuate quickly whereas low frequencies persist for up to 40 min. We observe that the measured fluctuations attenuate slowly, with the strongest frequency component reaching an amplitude of 10^{-2} after roughly 30 min. In the unfiltered and filtered models, this point is reached after 5 min and 15 min already. In addition, we observe a bandgap in the measurements and filtered model from 0.02 Hz to 0.1 Hz. The unfiltered model shows a gap up to 0.03 Hz.

The measured spectrogram shows harmonic components around 0 Hz, 0.4 Hz and 0.6 Hz, which are absent in both modelled spectrograms. As

these harmonics are present before and after the earthquake, we conjecture that they are caused by another external event, such as ocean waves.

Overall, the modelled and measured perturbed SOPs are qualitatively aligned.⁴ Both the shape of the spectrogram patterns, and the low-frequency bandgap, match. There is a mismatch in amplitude, which is modelled larger than measured at the start of the earthquake, and attenuates faster. This mismatch may be partially explained by the uncertainty in several model parameters. For example, the true values for h and ρ along the fiber, as well as the birefringence properties of the Curie cable are not known exactly. Moreover, our model assumes that changing water pressure is the primary SOP modulator as in [7]. However, [7] also notes that “for earthquake excitations [...], other mechanisms for SOP modulation cannot be excluded.” Besides uncertainty in the model parameters, the mismatch may therefore also reflect the presence of an additional coupling mechanism. In particular, as submarine cables partially lie on or in the seafloor, resulting longitudinal cable pulling may also add to the observed SOP modulation.

Conclusion

We presented a novel end-to-end model of earthquake-perturbed submarine fibre signal propagation. We verified the model on experimental measurements from the Curie cable during the 2020 Oaxaca earthquake. We believe that our model can aid large-scale design and testing of earthquake fibre sensing algorithms and help planning the placement of new sensing fibres.

The work presented here is a first step in the direction of a more complete open-source end-to-end modelling of earthquake sensing with optical fibres. Future work includes the inclusion of an analytical pressure filter [21], and the addition of SOP modulation by longitudinal fibre pulling.

⁴Normalised cross-correlation between the spectral centroids [23] of our filtered model and the measurements yields 0.70, showing a good quantitative match as well.

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